INCH-POUND

MIL-P-70524 (AR)
AMENDMENT 6
16 May 1996
SUPERSEDING
AMENDMENT 5
01 March 1996

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, XM913/XM927
WARHEAD INSULATION ASSEMBLY FOR

This amendment forms a part of Military Specification MIL-P-70524(AR), dated 29 October 1990, and is approved for use by the U.S. Army Armaments, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: change "XM913" to "M913" (and wherever it appears).

PAGE 3

3.5 Delete in its entirety and substitute the following:

"3.5 Proving ground. There shall be no warhead related premature explosive function, warhead breakup or separation of metal parts in the gun bore or in flight, when tested as specified in 4.5.8."

PAGE 4

3.7.1 Line 2, Delete "100 pounds" and substitute "2500 pounds".

3.7.2 Add the following new paragraph 3.7.2:

"3.7.2 Insulation bond. The insulator shall be completely and uniformly bonded to the warhead. No voids shall exist in the adhesive under the center 1 1/4 inch diameter of the insulator. Voids under the remaining area of the insulator shall not total more than 1/2 square inch with no single void in excess of 1/4 square inch."

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PAGE 7

4.3.1.2 Delete in its entirety.

PAGE 8

TABLE 1, Warhead insulation and assy. Delete "9341144" and substitute "9381144".

Warhead (prior to knurling and threading) Delete "4.4.2" and substitute "4.4.2.2".

Insulation, Bonding Assembly Pull Test:

Delete "25" and substitute "5".
Delete "4.5" and substitute "4.5.7".

PAGE 12

4.4.2.2 Special 1: Delete "3.9" and substitute "3.11".

PAGE 15

4.4.2.3 Major defect 118, Delete "Effective" and substitute "Thru".

Major defect 123, change "Surface finish of base thread" to "Surface finish of base thread undercut".

PAGE 17

4.4.2.4 Special defect 1 Delete "3.5" and substitute "3.6".
Special defect 2 Delete "3.7" and substitute "3.9".
Major defect 103 Delete "3.7" and substitute "3.9".

PAGE 21

4.4.2.7 Major defect 101, change "Insulation of adhesive above minimum recess: to "Insulation or adhesive above drawing requirement".

Major defect 102, delete in its entirety and replace with the following:

"102 Adhesive bond not complete and uniform 4.4.3.5 3.7.2 4.5.9".

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Add Major defect 103, as follows: "103, Bonding assembly pull test, 4.4.6, 3.7.1, 4.5.7"

PAGE 23

4.4.3.5 Add new paragraph 4.4.3.5 as follows:

"4.4.3.5 Insulation bond. This test shall be conducted 100% until 1205 consecutive assemblies comply with applicable requirements. After qualification, a sample of one assembly from each 7 shall be continuously selected and tested. Any sample assembly which fails to comply with applicable requirements shall be rejected. If a reject is found during sampling inspection, all units assembled during that shift shall be tested, and requalification of 1205 consecutive assemblies shall be required. If during 100% testing 2895 assemblies are tested without qualifying for sampling, the government shall be notified and corrective action shall be taken to improve the bonding process."

PAGE 24

Add New paragraphs 4.4.6, 4.4.6.1 and 4.4.6.2.

"4.4.6 Bonding assembly pull test. Sampling for the adhesive bond test of the Warhead insulator assembly consists of a process qualification and a normal sampling test. Testing shall be performed as specified in 4.5.7. The process qualification test shall be performed when any of the following conditions exists:

a. The start of production
b. A failure has occurred during normal sampling.
c. The lot of adhesive resin or hardener changes.
d. The bonding process (which includes surface preparation, mixing of adhesive, application of adhesive, fixturing and curing conditions) changes.
e. A lapse in production has exceeded one month.
f. As part of first article inspection.

4.4.6.1 Process qualification sampling. A random sample of five warhead bonding assemblies shall be taken from the initial day's production for testing. If any test sample fails to comply with the requirements of 3.7.1, the production quantity represented by that sample shall be rejected, the process corrected and an additional sample selected.

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4.4.6.2 Normal sampling. Once qualified per 4.4.6.1, from from each days bonding production, the contractor shall have one each assembly selected at random for testing. If any test sample fails to comply with the requirements of 3.7.1, the production quantity represented by that sample shall be rejected and the inspection sampling will revert back to 4.4.6.1 once corrective action has been taken on the process."

4.5.2 Delete "3.5.3" and substitute "3.6.3".

PAGE 28

4.5.6.2 Line 12, Delete "3.13" and substitute "3.8".

PAGE 29

4.5.7 Delete in its entirety and substitute the following:

"4.5.7 Bonding assembly pull test. Samples selected for this test shall be held until completion of the production adhesive cure cycle. Each sample shall have a pull test fixture bonded directly onto the exterior surface of the insulator (see 6.12). The sample assemblies shall be temperature conditioned at +145 degree F ± 5 degree F for 16 hours minimum prior to testing. Each test sample shall be pull (tension) tested within 15 minutes after removal from the conditioning chamber. The test fixture is to be designed in accordance with 4.4.4. Observation shall be made for compliance with requirements of 3.7.1. If the bond breaks at the fixture, it will be counted as a successful test providing the tension force is above the requirement of 3.7.1. If the bond breaks at the fixture with a lower tension value, it will be considered as a no test and another test sample must be selected. All test samples must have a new insulator bonded onto its base before returning it to the lot."

4.5.8 Delete in its entirety and substitute the following:

"4.5.8 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample warheads selected in accordance with 4.4.5 shall be TNT loaded, assembled with live rocket motors and delay assemblies, and conditioned to 145-5 degrees F for 16 hours minimum prior to firing rocket on in a M119 series howitzer using a PXR200 propelling charge conditioned to 70+/-2.5 degrees F. The warheads shall be fuzed with M577 series MTSQ fuzes set to 40 seconds minimum.

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shall be recorded for informational purposes only." Photographic and radar coverage shall be provided to assure compliance with 3.5. Range and deflection

4.5.9 Add new paragraph 4.5.9 as follows:

"4.5.9 Insulation bond test. This test shall utilize an infrared imaging system to detect voids in the adhesive bond under the insulator. The infrared system shall have the sensitivity to consistently detect and display the difference between areas of known bond and known voids of 1/8 inch diameter in a fabricated standard warhead. The standard shall be used for system check at the end of each shift's testing. If the standard voids are not detected, the shift's production shall be retested 100% or sampled, as applicable. The area under the insulator center boss cannot be tested. Apparent voids at the outside diameter under adhesive on the insulator aft surface shall be disregarded. Process control shall be utilized to minimize the amount of such adhesive buildup."

5.1 Delete in its entirety and substitute the following:

"5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-1169, Level C. Appropriate cushioning or partitions shall be provided to protect the items from contacting each other during shipment. Sufficient fillers must be used to insure a tight pack. Adequate corrosion protection shall be provided if required."

PAGE 30

6.6. Line 3, Delete "4.4.3.3" and substitute "4.4.5".

Line 5, Delete "3.6." and substitute "3.5".

PAGE 32

Add new paragraph 6.12

"6.12 Bonding assembly pull test - recommended fixture. Since there is not a convenient physical way to grasp the insulator, an adapter must be bonded to the insulator for the pull test. This requires that the insulator top surface be roughened in order to bond the adapter. The adapter must have its base slightly smaller than the insulator diameter. The adapter base must also be grooved and roughened to obtain a good bond. The top of the adapter must be threaded at its

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exact center to accept the pull test equipment fixturing. The insulator must be pulled squarely due to the peeling affects of the adhesive otherwise a lower pull value may be obtained. The epoxy that has been successfully used to bond the adapter is "DEVCON five minute epoxy". The assembly is allowed to cure from 1 to 3 hours before the warhead/insulator assembly is temperature conditioned. The above assembly is held in a Tinius Olsen (Tension tester) by a fixture that threads into the forward end of the warhead and the upper end is threaded into the fixture adapter. The tester is set at a cross head rate of 1"/min. with a full scale (12000 pounds) to perform test.

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